Banking-1A: Online Payment Fraud Detection, Progress Report

Break Through Tech AI Studio with Mastercard · Milestones 1 (September) and 2 (October) · Report prepared by Mohit Unecha for the Banking-1A team · October 20, 2026

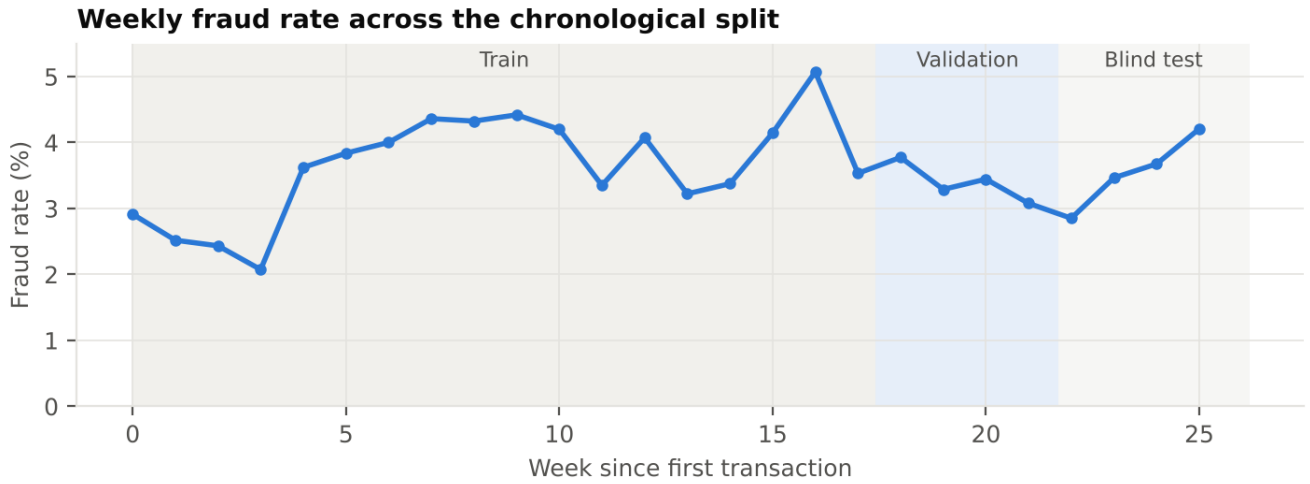
Summary

- **Data:** IEEE-CIS (Kaggle), 590,540 labeled card-not-present transactions, 3.5% fraud, 434 columns. Merged with the identity table and split by time: train 420,066 / validation 85,044 / blind test 85,430 (sealed until November).
- **Best model so far:** tuned XGBoost, **validation AUPRC 0.682 ± 0.0005** over 3 seeds (random = 0.034). Reviewing the riskiest 5% of transactions catches 71% of fraud.
- **Business result:** at a 1% daily review budget (about 28 reviews a day), ranking by **expected loss** (probability × amount) catches **48% of fraud dollars vs 21%** by probability alone. That's about $226k vs $110k saved over 30 days, using placeholder costs.
- **Deep learning:** the first MLP (focal loss) reaches AUPRC 0.536. As the literature predicts, it trails boosted trees on this data and doesn't yet help an ensemble.
- **Explainability:** link counts (cards, addresses and accounts tied to a payment) drive 60% of the reasons transactions get flagged. Missed fraud is mostly ordinary web purchases (product W) without identity data.

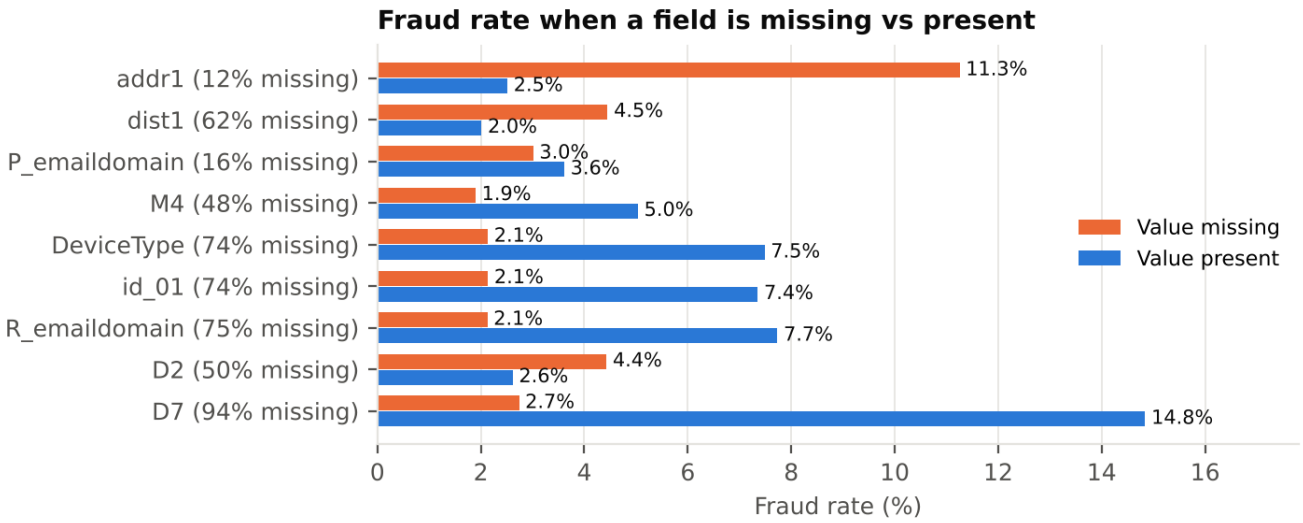
1. What we have done

Milestone	Work completed
September: data and baselines	Business objective and evaluation framework; literature and leakage review; data download, merge and canonical Parquet; frozen chronological split; EDA (missingness, categories, drift); customer-ID feasibility check; XGBoost and LightGBM baselines; categorical encoder; GitHub board and issues
October: tuning, deep learning, explainability	Optuna tuning for XGBoost (60 trials) and LightGBM; deep learning literature refresh; tabular MLP baseline; SHAP attribution; threshold calibration against the review budget; milestone review draft. In progress: CatBoost, logistic regression / random forest, FT-Transformer, customer-history features, ensemble, error analysis

2. Data findings



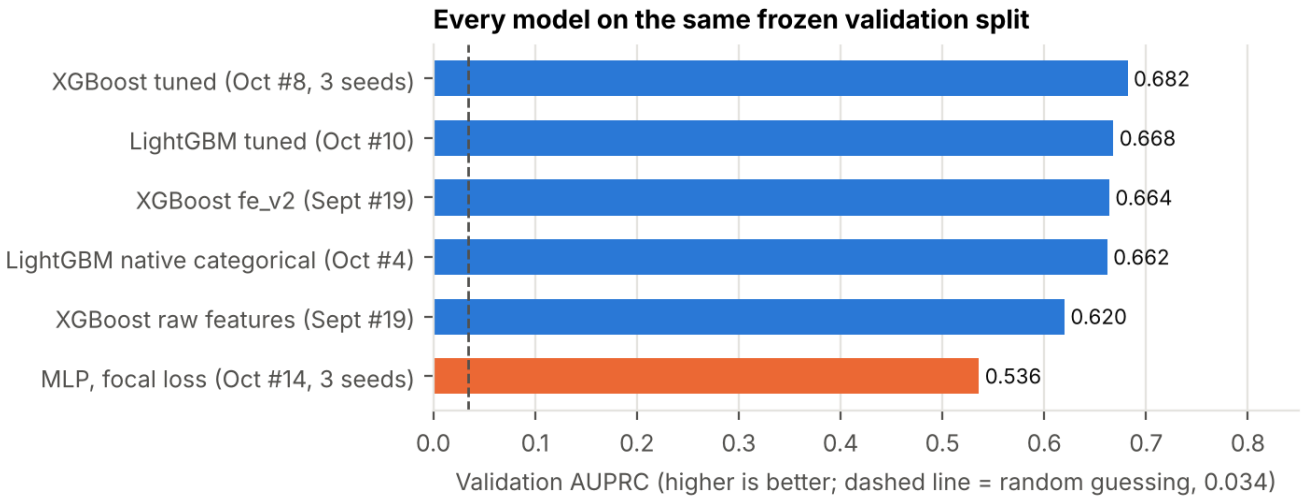
Fraud isn't stable over time: about 2.5% in the first month and 3.4% to 4.4% after (chi-square p < 1e-100), which is why every model is trained on earlier days and tested on later ones.



Missing values carry signal. A missing billing address goes with an 11.3% fraud rate vs 2.5% when present. Transactions with identity, device or recipient-email data are about three times riskier. Twenty features drift strongly between month 1 and month 6, led by browser version (id_31).

The 1st-place Kaggle customer ID (card1 + addr1 + day – D1) works on our data: the days since a customer's previous transaction match D3 95% of the time, vs 42% for card1 + addr1 alone. That makes leakage-safe customer-history features possible. It also shows fraud labels spread across a customer's transactions, so we must avoid letting the model simply recognize known bad customers.

3. Model results

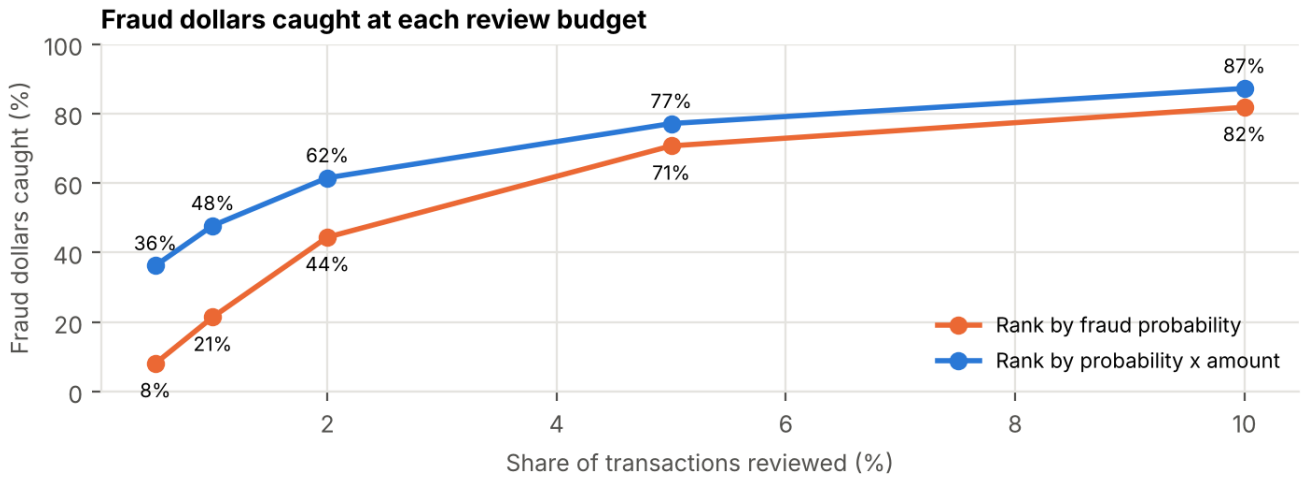


Model	AUPRC	ROC-AUC	Fraud caught, top 1%	Fraud caught, top 5%
XGBoost tuned (3 seeds)	0.682 ± 0.001	0.935	28.2%	71.2%
LightGBM tuned	0.668		28.1%	69.8%
XGBoost fe_v2	0.664	0.913	28.2%	69.8%
MLP, focal loss (3 seeds)	0.536 ± 0.008	0.906	25.2%	60.1%

The top 1% of validation is 851 transactions and validation has 2,884 frauds, so 29.5% is the most any model can catch at that rate. Tuning helped XGBoost by +0.018 AUPRC, 36 times the seed-to-seed spread, with deeper trees,

much stronger L1/L2 regularization, fewer categories per split and about half the full class weight.

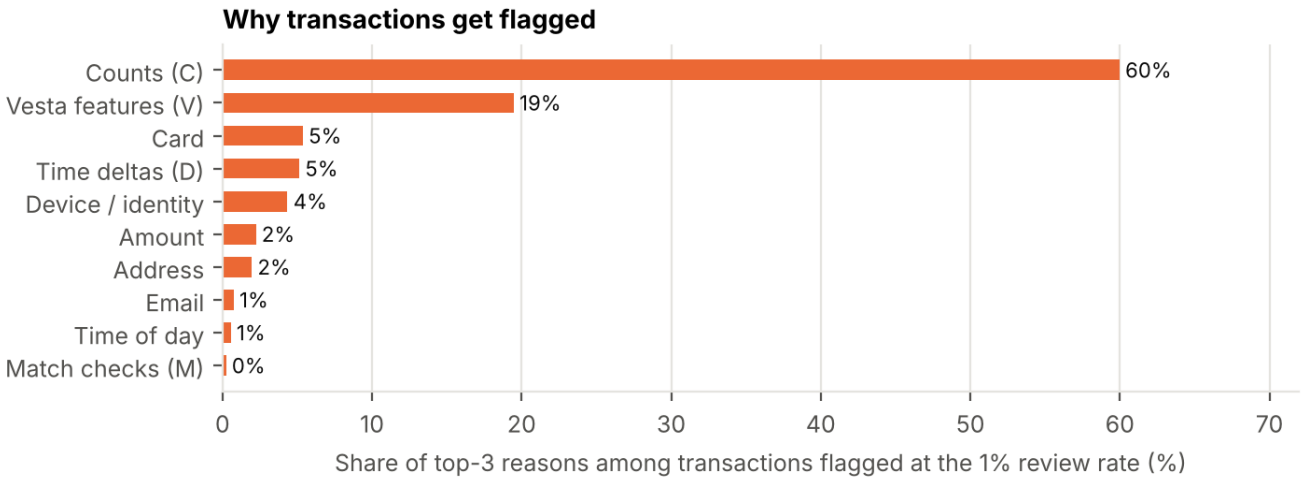
4. Operating the model: review budget and thresholds



1% review budget (28 a day)	Precision	Fraud cases caught	Fraud dollars caught	Net savings / 30 days
Rank by fraud probability	95%	28%	21% ($98k)	$110k
Rank by expected loss	65%	19%	**48% ($220k)**	**$226k**

Scores were calibrated with cross-fitted Platt scaling (the top decile predicts 27.1% fraud and observes 27.6%). The expected-loss threshold is stable across the validation month ($106 to $107). Fraud dollars caught at 5% fall from 83% to 71% between the first and second half of the month, which is the drift the November robustness work will address.

5. Why the model flags transactions (SHAP)



- About 75% of the model's influence comes from feature families an analyst can describe (counts, time deltas, card, address, amount, device, match checks, email). The other 24% sits in Vesta's anonymized V columns.
- Typical alert: "this payment is linked to an unusually large number of cards, addresses or accounts", a velocity pattern consistent with card testing and account takeover.
- Missed fraud (outside the top 5%) is mostly product W with no identity record (27% vs 59% for caught fraud), so customer-history features are the next lever.

6. Next steps (November)

- Finish October work in progress: CatBoost, logistic regression / random forest, FT-Transformer, customer-history features, ensemble, error analysis.
- Score the finalists once on the sealed blind test, then check robustness by week and segment.
- Build a small prototype that returns a fraud score, expected loss and the top three reasons.
- Get Challenge Advisor sign-off on costs ($20 chargeback, $5 review) and capacity (1% / 5% review rates).

Limitations: results are on validation, which was also used for tuning and thresholds, so they are slightly optimistic. Chargebacks arrive weeks late in real life, and most features are anonymized.